Week 1 report

Description and sources

	ONS vacancies	Indeed job postings
Series	AP2Y – UK Vacancies (thousands) – Total, all industries	indeed_job_postings_index_SA, "total postings", GB
Source	VACS01: Vacancies and unemployment	Indeed Hiring Lab job postings
Measures	Number of vacancies reported by employers that they are recruiting for.	Index of job postings on Indeed where 1 Feb 2020 = 100
License	Open Government License	Can be used by the public but need to cite Indeed Hiring Lab

ONS

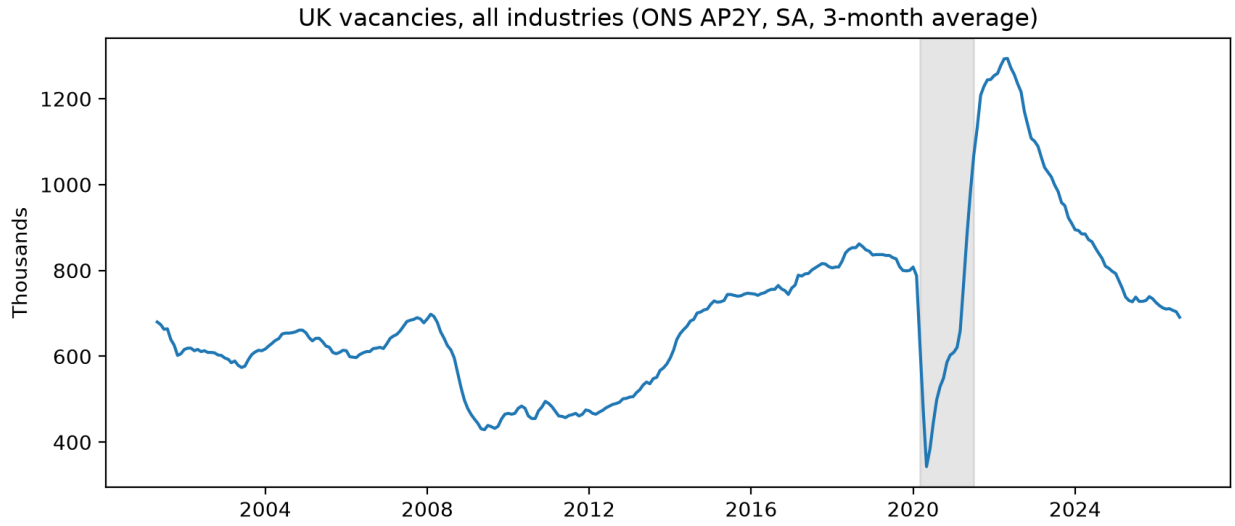
The Vacancy Survey asks about 6,100 businesses each month for their vacancies on the first Friday of the month. From the sample, 1,400 are large businesses which complete the survey every month and 4,700 are smaller businesses which complete the survey quarterly. This covers England, Wales and Scotland where Northern Ireland is estimated by weighting up the Great Britain data (Northern Ireland accounts for around 3% of UK employment), and includes all sectors apart from agriculture, forestry and fishing, households as employers and employment agencies to avoid double counting. The headline data series based on 3-month moving averages which are seasonally adjusted

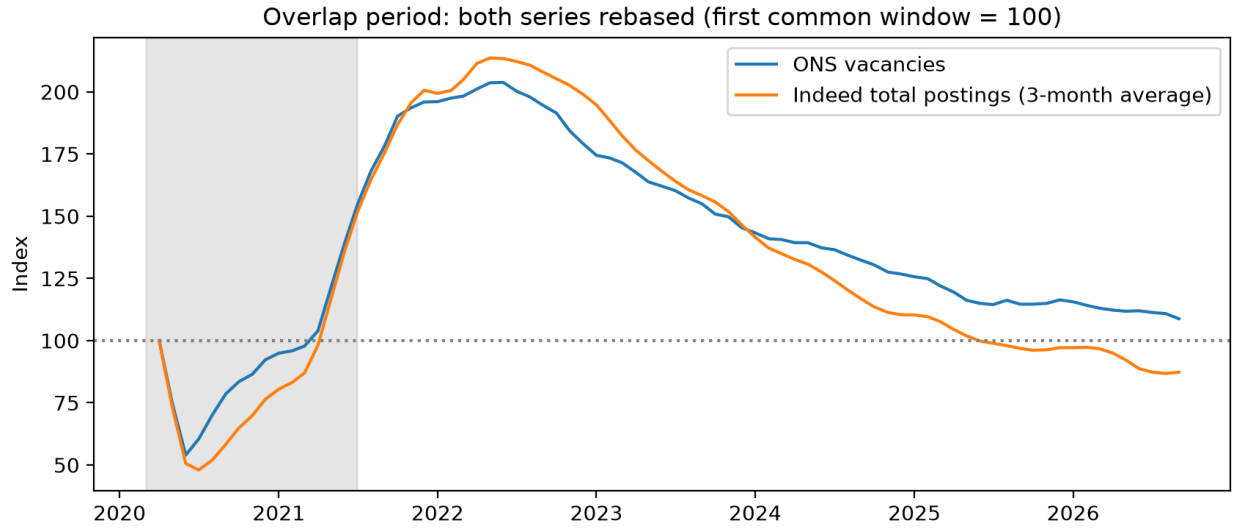
Indeed
Each daily value is a 7-day trailing average which shows the percentage change in seasonally adjusted job postings since 1 Feb 2020, which is the pre-pandemic baseline. The file contains the total job postings and new job postings (posted on Indeed for 7 or fewer than 7 days). We are using the total job postings as it matches the ONS vacancies at a point in time.

Historical Coverage and Observation Frequency

	ONS	**Indeed**
First Observation	Apr-Jun 2001	1 Feb 2020
Latest Observation	Jul-Sep 2026	16 Oct 2026
Number of observations	304	2450
Frequency	Monthly (3-month average)	Daily (7-day average)
Missing values or gaps?	None	None

Initial Findings and Visualisations





Trend

ONS vacancies rise and fall with the economy. They dropped after the 2008 financial crisis and recovered to just over 800 thousand by 2018. It then crashed to 343 thousand in May 2020 before increasing rapidly to 1.294 million in May 2022. Since then they have fallen steadily. Indeed job postings follow the same pattern.

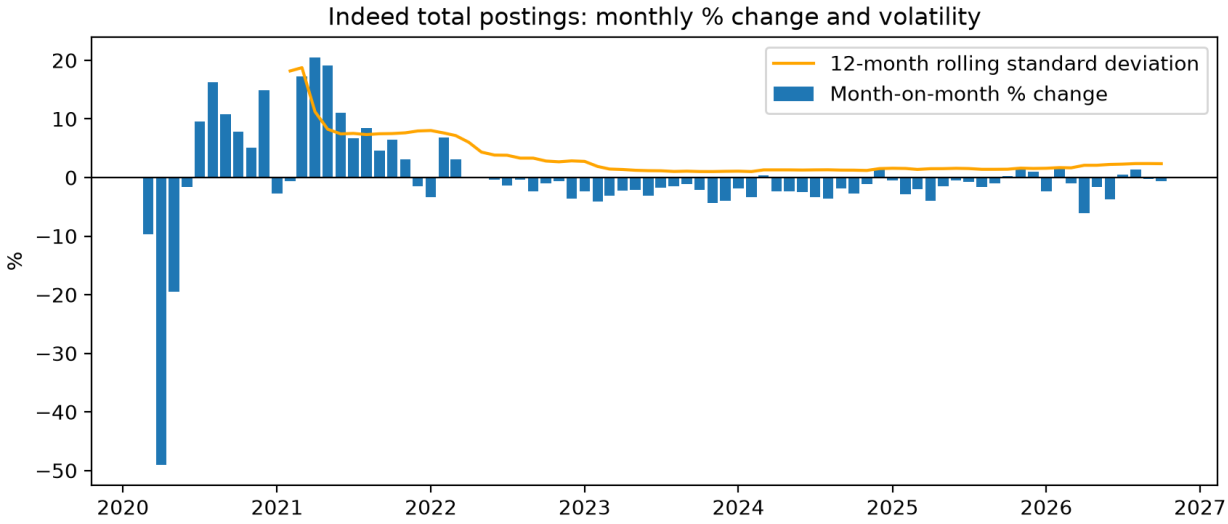
Seasonality

The ONS data is already seasonally adjusted, so any yearly patterns have been removed. Indeed publishes both seasonally-adjusted and non seasonally-adjusted figures. Comparing both of them shows that job postings are usually higher in September to October and lower in

December to January. We use the adjusted Indeed figures so that both series are treated in the same way.

Volatility

There are periods of unusually large changes, for example mainly during COVID. Before COVID, ONS vacancies usually changed by only about 1–2% a month. During COVID, the monthly changes were around ten times bigger. Since mid-2021 things have calmed down again. Indeed shows the same pattern and is slightly more jumpy overall.



Stationarity

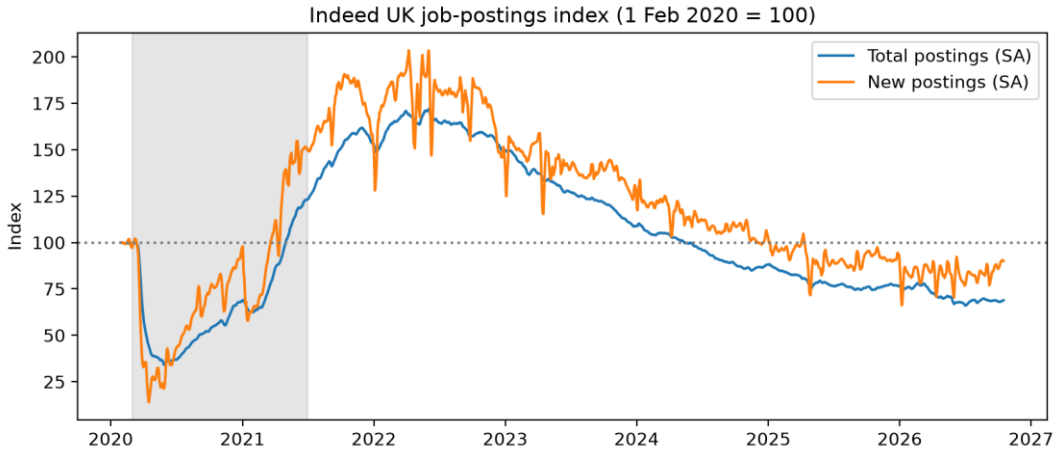
The average number of vacancies was about 635k before COVID and about 948k after, and the spread changes a lot over time too. If we look at the month-to-month changes instead, the average is much more stable (close to zero), although the changes were still far bigger during COVID. This suggests the models should work with monthly changes rather than raw values.

COVID-19

COVID is the biggest event in the overlapping period. It causes the largest drops and rises in both series, and Indeed's data starts just before it, on 1 February 2020. This means a large part of the data we can compare comes from a very unusual time.

Data-quality concerns and limitations

- No missing values or gaps in either dataset. The largest ONS monthly changes all fall in early 2020 and 2021 which show that these could reflect COVID rather than data errors
- ONS revisions: Each monthly release revises the previous 3 months and the same period a year earlier. In April of each year revisions of the previous 3 years are published.
- Indeed revisions: As new data arrives the seasonal adjustment is re-estimated and the whole history was revised when the methodology changed in November 2024.
- Different measurements: ONS is survey-based and counts the vacancies on one day each monthe where as Indeed is an index of job postings on a single platform. Since 2024 Indeed has fallen further than ONS so the relationship between their levels is not fixed.



- Sample: ONS excludes some sectors and estimates from Northern Ireland whereas Indeed covers all the roles advertised on its site.

Suitability for ARIMA vs ARIMAX

1. There is an overlap of 78 three-month windows from April 2020 to September 2026, where 63 are after June 2021. ARIMA can use the

225 months form the ONS data before Indeed starts, which ARIMAX cannot

2. Revisions in both series mean that the latest files are not what a forecaster saw at the time. The large COVID swings could also distort the models. Indeed also measures job postings on one website rather than official vacancies and it has decreased faster than ONS since 2024.
3. The datasets are matched up, ONS gives monthly three-month averages whereas Indeed is daily. Indeed can be turned into monthly averages and then three-month averages with the labels lined up so that both cover the same months. After they are matched the two series move closely together and their monthly changes have a correlation of 0.93.

Recommended priorities for Week 2:

- Build the cleaned and aligned monthly dataset
- Decide on how to treat COVID

Sources

- ONS VACS01: Vacancies and unemployment
- ONS AP2Y time series
- ONS Vacancy Survey QMI
- Indeed Hiring Lab job_postings_tracker